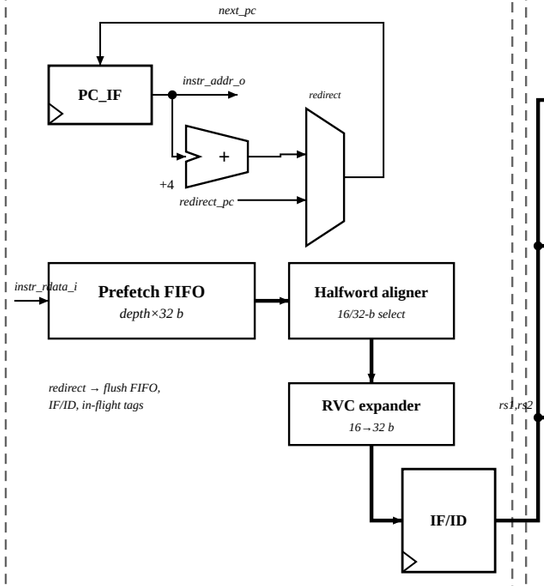
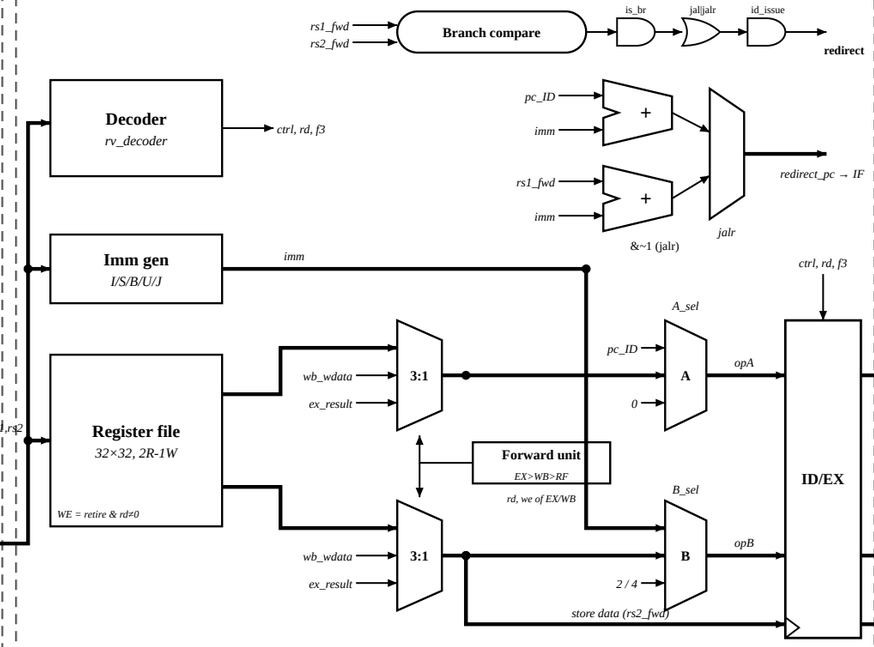
RV32IMC four-stage core – datapath / circuit level (no branch predictor). Bold = 32-bit bus, thin = control.

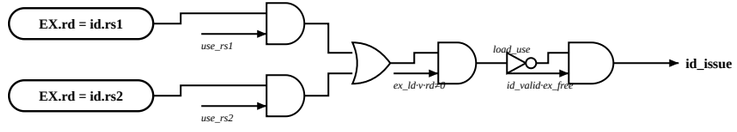
IF – Instruction Fetch



ID – Decode / Register Read / Branch Resolve



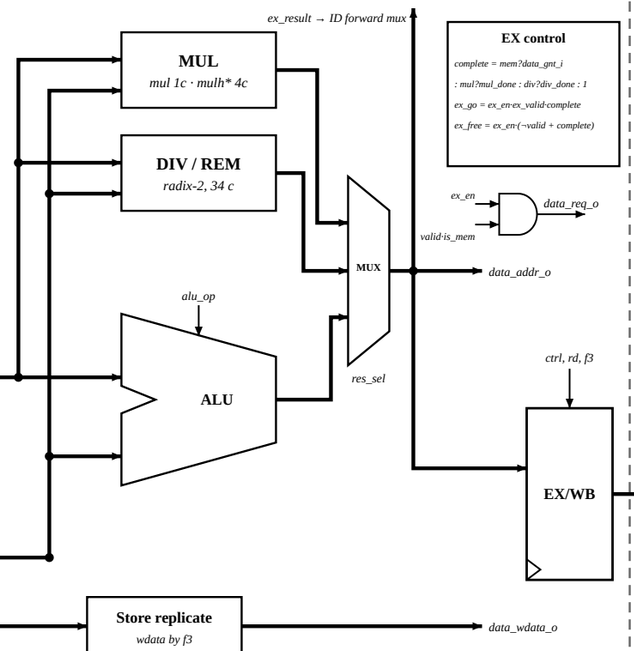
Hazard / issue control



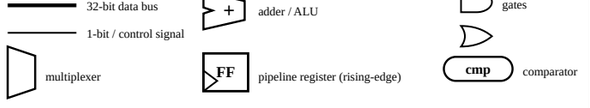
load_use = id_valid · ex_valid · ex_load · (EX.rd=rs1·use_rs1 + EX.rd=rs2·use_rs2)

id_issue = id_valid · ¬load_use · ex_free ; redirect = id_issue · (jal + jalr + branch·taken)

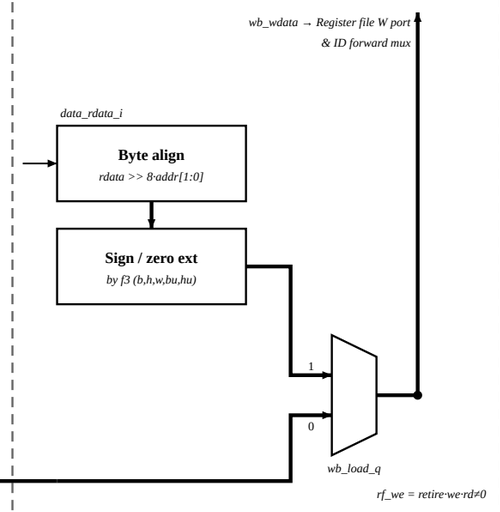
EX – Execute



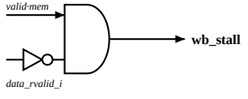
Legend



WB – Write-back



WB control



ex_en = ¬wb_stall (freezes IF/ID/EX/WB)

retire = wb_valid_q · ¬wb_stall

load/store complete only on data_rvalid_i